1. ปรับปรุงข้อมูล Received JSON ที่ได้รับมา(ด้านล่าง) ให้อยู่ในรูปแบบข้อมูล Results JSON ที่ต้องการ.
*สามารถเขียนด้วยภาษา หรือ Framework ใดก็ได้ที่ถนัด หรือ ถ้าจะให้ตรงกับงาน เขียนด้วยภาษา Python สามารถใช้ lib ได้เช่น numpy pandas เป็นต้น

Received JSON (ข้อมูล JSON ที่ได้รับมา)


```
{
	"students": [{
				"user_id": 1,
				"first_name": "Dorree",
				"last_name": "Stanlick",
				"gender_id": "a2",
				"adore_car": "c4",
				"car_brand": [
					"c1",
					"c4"
				],
				"color": [
					"IndianRed",
					"LightCoral"
				]
			},
			{
				"user_id": 2,
				"first_name": "Gert",
				"last_name": "Fake",
				"gender_id": "a1",
				"adore_car": "c1",
				"car_brand": [
					"c1",
					"c2"
				],
```

```
        "color": [
            "DarkSalmon",
            "LightSalmon"
        ]
    },
    {
        "user_id": 3,
        "first_name": "Rene",
        "last_name": "Dust",
        "gender_id": "a2",
        "adore_car": "c5",
        "car_brand": [
            "c3",
            "c5"
        ],
        "color": [
            "Salmon",
            "LightCoral"
        ]
    },
    {
        "user_id": 4,
        "first_name": "Clo",
        "last_name": "Cordes",
        "gender_id": "a1",
        "adore_car": "c4",
        "car_brand": [
            "c2",
            "c4",
            "c5"
```

```
				],
				"color": [
					"DarkSalmon"
				]
			},
			{
				"user_id": 5,
				"first_name": "Emlynne",
				"last_name": "Kettley",
				"gender_id": "a2",
				"adore_car": "c1",
				"car_brand": [
					"c1"
				],
				"color": [
					"IndianRed",
					"LightSalmon",
					"LightCoral",
					"Salmon"
				]
			}
		],
		"gender": [{
				"gender_id": "a1",
				"label": "male"
			},
			{
				"gender_id": "a2",
				"label": "female"
			}
```

],
"cars": [{
"car_id": "c1",
"car_brand": "Subaru",
"car_make": "Japan"
},
{
"car_id": "c2",
"car_brand": "Dodge",
"car_make": "United States"

},
{
"car_id": "c3",
"car_brand": "Volkswagen",
"car_make": "Germany"

},
{
"car_id": "c4",
"car_brand": "Hyundai",
"car_make": "Korea"

},
{
"car_id": "c5",
"car_brand": "Toyota",
"car_make": "Japan"

}

```
],

"rgbCode": [{
			"hex": "#CD5C5C",
			"label": "IndianRed",
			"rgb": "205, 92, 92"
		},
		{
			"hex": "#F08080",
			"label": "LightCoral",
			"rgb": "240, 128, 128"
		},
		{
			"hex": "#FA8072",
			"label": "Salmon",
			"rgb": "250, 128, 114"
		},
		{
			"hex": "#E9967A",
			"label": "DarkSalmon",
			"rgb": "233, 150, 122"
		},
		{
			"hex": "#FFA07A",
			"label": "LightSalmon",
			"rgb": "255, 160, 122"
		}
],
"countries": [{
			"label": "Czech Republic",
```

"user_ids": [{
"userId": "1"
},
{
"userId": "3"
},
{
"userId": "5"
}
]
},
{
"label": "Colombia",
"user_ids": [{
"userId": "2"
},
{
"userId": "4"
}
]
}
]
}

Results JSON (ผลลัพธ์ JSON ที่ต้องการ)

[
{

```
    "user_id": 1,
    "first_name": "Dorree",
    "last_name": "Stanlick",
    "gender": "female",
    "adore_car" : {
        "car_brand": "Hyundai",
              "brand_from": "Korea"
          },
    "car_brand": ["Subaru", "Hyundai"],
    "countries": "Czech Republic",
    "colors": [
        {
            "hex": "#CD5C5C",
            "label": "IndianRed",
            "rgb": "205, 92, 92"
        },
        {
            "hex": "#F08080",
            "label": "LightCoral",
            "rgb": "240, 128, 128"
        }
    ]
},
{
    "user_id": 2,
    "first_name": "Gert",
    "last_name": "Fake,
    "gender_id": "male",
    "adore_car" : {
        "car_brand": "Subaru",
```

"brand_from": "Japan"
},
"car_brand": ["Subaru", "Dodge"],
"countries": "Colombia",
"colors": [
{
"hex": "#E9967A",
"label": "DarkSalmon",
"rgb": "233, 150, 122"
},
{
"hex": "#FFA07A",
"label": "LightSalmon",
"rgb": "255, 160, 122"
}
]
},

... **และข้อมูล ผู้ใช้ที่เหลือทั้งหมด**

]

2. เขียนฟังก์ชัน cosine_similarity(vec1: List[float], vec2: List[float]) -> float

เพื่อคำนวณ cosine similarity ระหว่างเวกเตอร์สองตัว ด้วยภาษา Python

vec1 = [1, 2, 3]

vec2 = [2, 3, 4]

print(cosine_similarity(vec1, vec2))

OUTPUT

ANS => 0.9926

3. Sentiment Classification Pipeline

คำอธิบาย:

ให้ผู้สมัครสร้าง Mini Machine Learning Pipeline สำหรับการทำ Sentiment Classification (positive/negative) โดยใช้ Python และ scikit-learn

dataset : https://lief-assets-storage.sgp1.cdn.digitaloceanspaces.com/Test/dataset.txt

สิ่งที่ต้องทำ

Preprocessing ข้อความ

แปลงข้อความเป็นตัวพิมพ์เล็กทั้งหมด

ลบ punctuation และ stopwords (["is", "the", "and", "a", "an", "of"])

แยกคำ (tokenization)

Feature Engineering

แปลงข้อความเป็น Bag-of-Words หรือ TF-IDF vector (ใช้ scikit-learn ได้)

Model Training

ใช้ Logistic Regression ในการ train โมเดล

แบ่ง dataset เป็น train 80% / test 20%

Evaluation

แสดงผล accuracy ของโมเดลบน test set

แสดง confusion matrix

ข้อมูล Dataset

label = 1 หมายถึง Positive, label = 0 หมายถึง Negative